

Challenge (Habanero)

- Q1.** In a hike, the height h of a mountain in miles above sea level is given by $h = -0.5t^2 + 2t + 10$, where t is time in hours. What is the discriminant of this equation?
- (A) -4
(B) 4
(C) 0
(D) 16
(E) -16
- Q2.** The population P of rabbits in a forest is modeled by $P = 2t^2 + 5t + 20$, where t is time in months. How many solutions does this equation have when $P = 0$?
- (A) 1
(B) 0
(C) 2
(D) Real solutions do not exist
(E) Cannot be determined
- Q3.** A bus travels from one city to another with a speed s given by $s = -t^2 + 4t + 5$, where t is time in hours. If the bus's speed is 0 , how many real solutions does this equation have?
- (A) 1
(B) 2
(C) 0
(D) Real solutions do not exist
(E) Cannot be determined
- Q4.** For a quadratic equation $at^2 + bt + c = 0$, if $a = 1$ and $b = -3$, and the discriminant is 5 , what is the value of c ?
- (A) -4
(B) 1
(C) -1
(D) 2
(E) 4
- Q5.** A park ranger travels through the forest at a rate of r miles per hour given by $r = -2t^2 + t + 6$, where t is time in hours. If $r = 0$, what is the value of the discriminant of this equation?
- (A) -23
(B) 23
(C) -11
(D) 11
(E) 1
- Q6.** The height h of a rock thrown upward is modeled by $h = -4t^2 + 20t + 1$, where t is time in seconds. How many real solutions does this equation have when $h = 0$?
- (A) 2
(B) 1
(C) 0
(D) Real solutions do not exist
(E) Cannot be determined
- Q7.** A bird researcher tracks the population P of birds in a sanctuary given by $P = 3t^2 - 2t - 1$, where t is time in months. What is the discriminant of this equation?
- (A) 25
(B) 19
(C) 13
(D) 1
(E) 4
- Q8.** The speed s of a bicycle is given by $s = t^2 + 2t - 3$, where t is time in hours. If $s = 0$, what is the nature of the solutions of this equation?
- (A) Two distinct real solutions
(B) One real solution
(C) No real solutions
(D) Imaginary solutions
(E) Cannot be determined

Open Ended Questions (FRQ)

- Q9.** For a quadratic equation $x^2 + bx + c = 0$, if the discriminant is 45 , and one of the roots is -2 , find the values of b and c . Show your work and explain your reasoning.
- Q10.** A group of hikers are at an elevation h given by $h = -t^2 + 5t + 12$, where t is the time in hours. Find the time t when $h = 0$ by using the quadratic formula. Show your calculations and give a brief explanation of your method.

Chef's Tips

- Ensure students have all necessary materials before starting the test
- Watch for students who may need extra time to complete the test
- Be available to answer questions about the test format but not the content